

K Nearest Neighbors

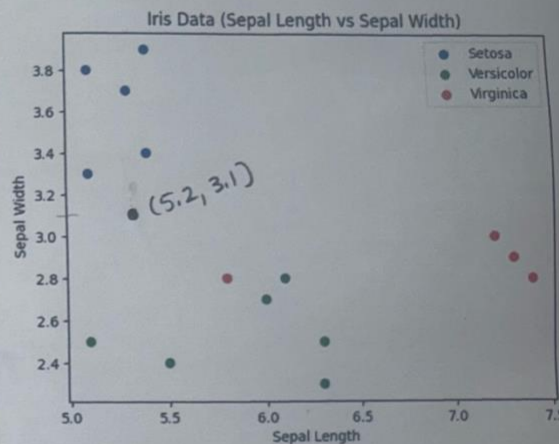
Given the following sepal data, classify its species: Length = 5.2, Width = 3.1

Step 1) Find the Euclidean distance from the test point to each point in the sample

Step 2) Rank each point from smallest to largest

Step 3) Use the given k to find the species based on the majority

Length	Width	Species	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Rank	k = 1	k = 3	k = 5
5.3	3.7	Setosa	$\sqrt{(5.2-5.3)^2 + (3.1-3.7)^2} = 0.606$	3		✓	✓
5.1	3.8	Setosa	$\sqrt{(5.2-5.1)^2 + (3.1-3.8)^2} = 0.707$	6			
7.2	3.0	Virginica	$\sqrt{(5.2-7.2)^2 + (3.1-3.0)^2} = 2.002$	13			
5.4	3.4	Setosa	$\sqrt{(5.2-5.4)^2 + (3.1-3.4)^2} = 0.361$	2		✓	✓
5.1	3.3	Setosa	$\sqrt{(5.2-5.1)^2 + (3.1-3.3)^2} = 0.224$	1	✓	✓	✓
5.4	3.9	Setosa	$\sqrt{(5.2-5.4)^2 + (3.1-3.9)^2} = 0.825$	8			
7.4	2.8	Virginica	$\sqrt{(5.2-7.4)^2 + (3.1-2.8)^2} = 2.220$	15			
6.1	2.8	Versicolor	$\sqrt{(5.2-6.1)^2 + (3.1-2.8)^2} = 0.949$	10			
7.3	2.9	Virginica	$\sqrt{(5.2-7.3)^2 + (3.1-2.9)^2} = 2.110$	14			
6.0	2.7	Versicolor	$\sqrt{(5.2-6.0)^2 + (3.1-2.7)^2} = 0.894$	9			
5.8	2.8	Virginica	$\sqrt{(5.2-5.8)^2 + (3.1-2.8)^2} = 0.671$	5			✓
6.3	2.3	Versicolor	$\sqrt{(5.2-6.3)^2 + (3.1-2.3)^2} = 1.360$	12			
5.1	2.5	Versicolor	$\sqrt{(5.2-5.1)^2 + (3.1-2.5)^2} = 0.608$	4			✓
6.3	2.5	Versicolor	$\sqrt{(5.2-6.3)^2 + (3.1-2.5)^2} = 1.253$	11			
5.5	2.4	Versicolor	$\sqrt{(5.2-5.5)^2 + (3.1-2.4)^2} = 0.762$	7			



k=1 setosa

k=3: setosa
setosa
setosa

k=5: setosa
setosa
setosa
Versicolor
Virginica